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Indene formation upon borane-induced cyclization of arylallenes, 1,1-carboboration, and retro-hydroboration†

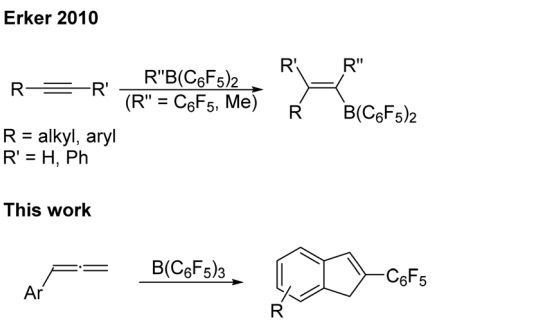
Max Hasenbeck,[a] Felix Wech,[a] Arthur Averdunk,[a] Jonathan Becker[b] and Urs Gellrich *[a]

We herein report the reaction of arylallenes with tris(pentafluorophenyl)borane that yields pentafluorophenyl substituted indenes. The tris(pentafluorophenyl)borane induces the cyclization of the allene and transfers a pentafluorophenyl ring in the course of this reaction. A Hammett plot analysis and DFT computations indicate a 1,1-carboboration to be the C–C bond-forming step.

In 2010, Erker and co-workers reported that the reaction of the strongly Lewis acidic tris(pentafluorophenyl)borane ($B(C_6F_5)_3$) with terminal alkynes leads to a 1,1-carboboration of the alkyne.[1] The substrate scope of the 1,1-carboboration was extended to phosphinyl substituted alkynes, yielding alkenyl-bridged frustrated Lewis pairs.[2] Furthermore, the 1,1-carboboration of internal alkynes with concomitant C–C cleavage was demonstrated.[3,4] More recently, Melen *et al.* reported the 1,2-carboboration of allenyl ketones.[5]

We found now that the reaction of $B(C_6F_5)_3$ with phenylallene **1** leads to the formation of the pentafluorophenyl-substituted indene **2** (Scheme 1).[6] Our initial finding was that the addition of **1** to a solution of $B(C_6F_5)_3$ in dichloromethane-d_2 (DCM-d_2) leads to the formation of indene **2** in 48% NMR-yield within 45 minutes (Scheme 2).

Hence, the $B(C_6F_5)_3$ induces a ring closure and transfers a pentafluorophenyl ring to the allene.[7] As a side product of this reaction, we observed the allylborane **3** that is likely formed by hydroboration of the phenylallene **1**.[8] Besides, minor amounts of a vinylborane, originating from the hydroboration of the internal double bond, and a 2-boryl-hexa-1,5-dien, formed by Piers' borane induced dimerization of the allene, were detected (for details, see the ESI†). The detection of **3** gives a direct hint to the intermediate formation of Piers' borane $HB(C_6F_5)_2$ in the course of the reaction.[9] Accordingly, the reaction constitutes a formal exchange of a pentafluorophenyl group of the $B(C_6F_5)_3$ with hydrogen. The reaction was subsequently performed on a preparative scale and the indene **2** was isolated and fully characterized. The structural assignment is further supported by SCXRD (Scheme 2). We then investigated the scope of this transformation by reacting different aryl allenes with $B(C_6F_5)_3$. As the hydroboration observed in the initial experiment consumes some of the allene, these reactions were performed with two equivalents of the allene. Under these conditions, **2** was isolated in 68% yield with respect to $B(C_6F_5)_3$, the limiting reagent (Scheme 3). The same reaction with one equivalent phenylallene and norbornene as sacrificial olefine to capture Piers' borane yielded **2** in 58% yield. The benzindene **4** is obtained in 65% yield from the reaction of 1-naphthylallene with $B(C_6F_5)_3$. However, it was isolated as a mixture of isomers that differ regarding the position of the benzylic double bond. Likewise, the reaction of *p*-tolylallene and *p*-(isopropyl)phenylallene gives the indenes **5** and **6** as a mixture of regioisomers.



Scheme 1 The context of this work: the 1,1-carboboration of alkynes described by Erker and the borane-induced ring closure with concomitant aryl group transfer described herein.

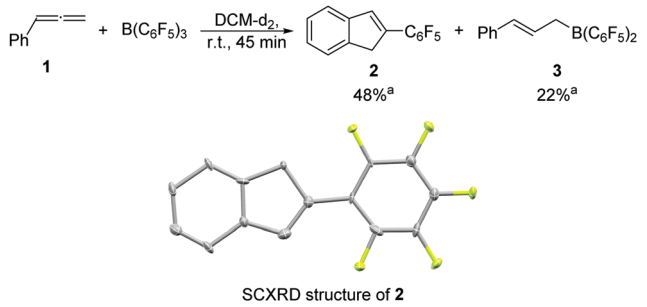
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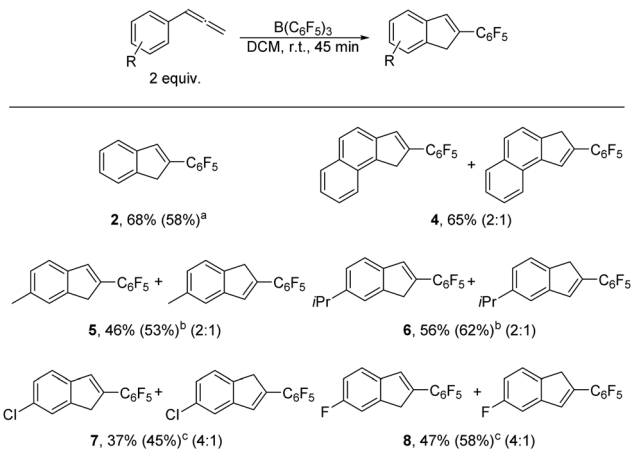
† Electronic supplementary information (ESI) available: Experimental and computational details, spectroscopic and crystallographic data, copies of NMR spectra. CCDC 2059168. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d1cc01750k

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Scheme 2 Formation of the pentafluorophenyl-substituted indene **2** upon the reaction of **1** with $B(C_6F_5)_3$. [a]Yields determined by qNMR with 1,3,5-trimethoxybenzene as internal standard. The inset shows the molecular structure of **2** derived from SCXRD (50% probability ellipsoids, all hydrogen atoms are omitted for clarity).



Scheme 3 Scope of the $B(C_6F_5)_3$ mediated formation of indenes from allenes. Yields of products isolated by column chromatography are given. (a) With one equivalent phenylallene and one equivalent norbornene. (b) Reaction temperature 0 °C. (c) Oil bath temperature 60 °C.

A better yield was obtained when the reactions with these substrates were run at 0 °C, presumably because of side reactions at r.t. In contrast, the reactions with *p*-(chloro)phenylallene and *p*-(fluoro)phenylallene required an elevated reaction temperature to give **7** and **8** in moderate yields. These findings indicate that electron-rich arylallenes undergo a faster cyclization whereas electron-withdrawing substituents slow down the reaction.

The effect of the substituents on the phenyl ring on the reaction rate was further assessed by a Hammett analysis (Fig. 1).[10] The negative slope of the Hammett plot and the rho value of -3.9 ± 0.7 indicate that in the rate-determining transition state of this reaction positive charge is built up in the benzylic position of the arylallene.

The increased reactivity of electron-rich allenes was used for a cyclization with a less Lewis acidic borane. The reaction of *p*-(isopropyl)phenylallene with $MeB(C_6F_5)_2$ leads to cyclization of the allene and transfer of the methyl group (Scheme 4). However, the formation of methylindene **9** required a prolonged reaction time.

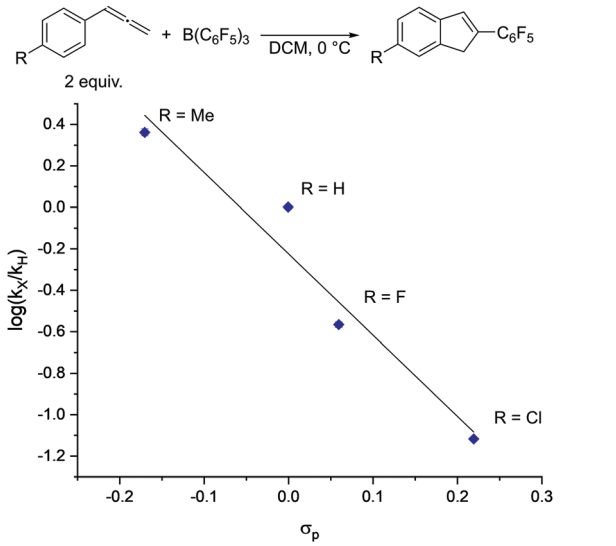
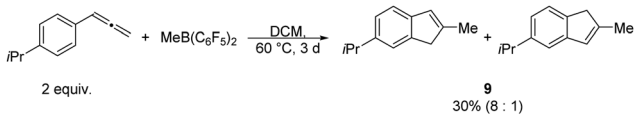


Fig. 1 Hammett plot analysis of the $B(C_6F_5)_3$- mediated formation of indenes from aryl allenes. Note that only the major isomer is shown in the scheme.



Scheme 4 Formation of the methyl indene 9 upon reaction of *p*-(isopropyl)phenylallene with $MeB(C_6F_5)_2$.

The mechanism of the $B(C_6F_5)_3$-mediated indene formation was further investigated computationally by DFT at the revDSD-PBEP86-D4/def2-QZVPP//PBEh-3c level of theory (Fig. 2).[11–13] As zwitterionic intermediates are likely involved in the reaction, the SMD solvent model for DCM was used for the structure optimizations and the single point computations.[14] We assume that the reaction commences with the addition of $B(C_6F_5)_3$ to the π-system of **1**. This step yields the zwitterion **INT-1**. The positive charge in this intermediate is stabilized by allyl and benzyl resonance. An intramolecular Friedel–Crafts alkylation *via* **TS-2** closes the five-membered ring of the indene core. We further assume that an intermolecular proton shift leads to a re-aromatization and the intermediate **INT-3** (Scheme 5). However, all attempts to optimize the structure of **INT-3** resulted in a pentafluorophenyl transfer and lead to **INT-4**. Relaxed potential energy surface scans further showed that the pentafluorophenyl transfer is a barrierless process. This finding agrees with computational studies of the 1,1-carboboration of alkynes by Erker, Grimme, and co-workers that showed that once a carbocation is formed in α-position to the $B(C_6F_5)_3$ moiety, the pentafluorophenyl transfer is barrierless.

A retro-hydroboration, *i.e.* the liberation of $HB(C_6F_5)_2$ from **Int-4**, forms product **2**. According to the computations, this step requires only a moderate activation energy of 16.7 kcal mol^{-1}. Notably, the retro-hydroboration is computed to be exergonic by 2.2 kcal mol^{-1}. According to the computations, **TS-1**, the

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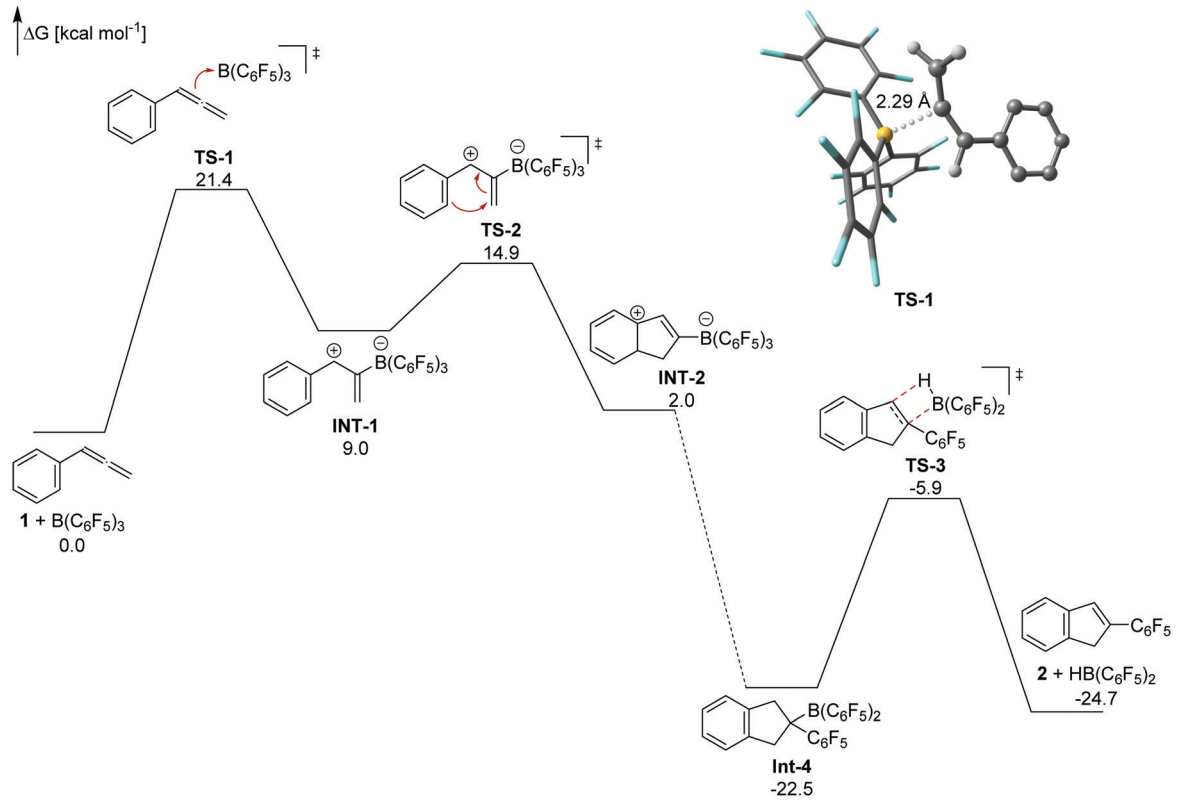
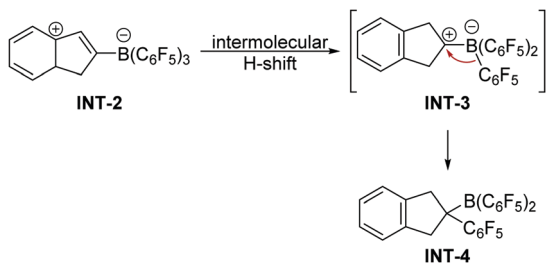


Fig. 2 Gibbs free energies of the $B(C_6F_5)_3$-mediated indene formation computed at the revDSD-PBEP86-D4/def2-QZVPP//PBEh-3c level of theory. The SMD solvent model for DCM was used for the structure optimizations and the single point computations. The inset shows the optimized structure of **TS-1**.



Scheme 5 Proposed mechanism for the pentafluorophenyl transfer.

addition of the phenyl allene to $B(C_6F_5)_3$, is the rate-determining transition state. In this transition state, a positive charge is built up in the benzylic position that is stabilized by electron-donating groups in the para position. Thus, the computations are in favorable agreement with the Hammett analysis. In summary, we have documented that the reaction of arylallenes with $B(C_6F_5)_3$ leads to pentafluorophenyl substituted indenes. A plausible mechanistic scenario consists of a $B(C_6F_5)_3$-induced cyclization, a pentafluorophenyl transfer, and a retro-hydroboration. This reaction is a rare example of a metal-free C_{sp^2}–C_{sp^2} bond formation.[15] The results reported herein might inspire the development of synthetic methods that rely on 1,1-carboboration and retro-hydroboration reactions.

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Conflicts of interest

There are no conflicts to declare.

Notes and references

1 (*a*) C. Chen, F. Eweiner, B. Wibbeling, R. Fröhlich, S. Senda, Y. Ohki, K. Tatsumi, S. Grimme, G. Kehr and G. Erker, *Chem. – Asian J.*, 2010, **5**, 2199–2208; (*b*) C. Chen, T. Voss, R. Fröhlich, G. Kehr and G. Erker, *Org. Lett.*, 2011, **13**, 62–65; (*c*) B.-H. Xu, G. Kehr, R. Fröhlich, B. Wibbeling, B. Schirmer, S. Grimme and G. Erker, *Angew. Chem., Int. Ed.*, 2011, **50**, 7183–7186; (*d*) R. Liedtke, R. Fröhlich, G. Kehr and G. Erker, *Organometallics*, 2011, **30**, 5222–5232; (*e*) C. Chen, R. Fröhlich, G. Kehr and G. Erker, *Chem. Commun.*, 2010, **46**, 3580–3582.
2 (*a*) O. Ekkert, R. Fröhlich, G. Kehr and G. Erker, *J. Am. Chem. Soc.*, 2011, **133**, 4610–4616; (*b*) O. Ekkert, G. Kehr, R. Fröhlich and G. Erker, *Chem. Commun.*, 2011, **47**, 10482–10484.
3 (*a*) B.-H. Xu, G. Kehr, R. Fröhlich, S. Grimme and G. Erker, *J. Am. Chem. Soc.*, 2011, **113**, 3480–3491; (*b*) C. Chen, G. Kehr, R. Fröhlich and G. Erker, *J. Am. Chem. Soc.*, 2010, **132**, 13594–13595.
4 For a recent review on 1,1-carboboration reactions see: G. Kehr and G. Erker, *Chem. Commun.*, 2012, **48**, 1839–1850.
5 R. L. Melen, L. C. Wilkins, B. M. Kariuki, H. Wadepohl, L. H. Gade, A. S. K. Hashmi, D. W. Stephan and M. M. Hansmann, *Organometallics*, 2015, **34**, 4127–4137.
6 Upon hydrogenation of tetraphenylallene with a frustrated Lewis pair consisting of $PhNMe_2$ and $B(C_6F_5)_3$ Alcarazo and co-workers observed cyclization of the allene as side reaction. However, no

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carboboration was reported: B. Inés, D. Palomas, S. Holle, S. Steinberg, J. A. Nicasio and M. Alcarazo, *Angew. Chem., Int. Ed.*, 2012, **51**, 12367–12369.

7 The formation of 2-borylindenes upon the reaction of arylallenes with $ClB(C_6F_5)_2$ and $BrB(C_6F_5)_2$ was reported: X. Tao, C. G. Daniliuc, K. Soloviova, C. A. Strassert, G. Kehra and G. Erker, *Chem. Commun.*, 2019, **55**, 10166–10169.

8 (*a*) X. Tao, C. G. Daniliuc, D. Dittrich, G. Kehr and G. Erker, *Angew. Chem., Int. Ed.*, 2018, **57**, 13922–13926; (*b*) M. Hasenbeck, S. Ahles, A. Averdunk and U. Gellrich, *Angew. Chem., Int. Ed.*, 2020, **59**, 23885–23891.

9 (*a*) D. J. Parks, R. E. von, H. Spence and W. E. Piers, *Angew. Chem., Int. Ed. Engl.*, 1995, **34**, 809–811; (*b*) E. A. Patrick and W. E. Piers, *Chem. Commun.*, 2020, **56**, 841–853.

10 L. P. Hammett, *J. Am. Chem. Soc.*, 1937, **59**, 96–103.

11 (*a*) G. Santra, N. Sylvetsky and J. M. L. Martin, *J. Phys. Chem. A*, 2019, **123**, 5129–5143; (*b*) E. Caldeweyher, C. Bannwarth and S. Grimme, *J. Chem. Phys.*, 2017, **147**, 034112; (*c*) E. Caldeweyher, S. Ehlert, A. Hansen, H. Neugebauer, S. Spicher, C. Bannwarth and S. Grimme, *J. Chem. Phys.*, 2019, **150**, 154122; (*d*) F. Weigend and R. Ahlrichs, *Phys. Chem. Chem. Phys.*, 2005, **7**, 3297–3305; (*e*) A. Hellweg, C. Hattig, S. Hofener and W. Klopper, *Theor. Chem. Acc.*, 2007, **117**, 587–597; (*f*) F. Weigend, *J. Comput. Chem.*, 2008, **29**, 167–175.

12 (*a*) S. Grimme, J. G. Brandenburg, C. Bannwarth and A. Hansen, *J. Chem. Phys.*, 2015, **143**, 054107; (*b*) H. Kruse and S. Grimme, *J. Chem. Phys.*, 2012, **136**, 154101; (*c*) S. Grimme, S. Ehrlich and L. Goerigk, *J. Comput. Chem.*, 2011, **32**, 1456–1465; (*d*) S. Grimme, J. Antony, S. Ehrlich and H. Krieg, *J. Chem. Phys.*, 2010, **132**, 154104; (*e*) F. Weigend, *Phys. Chem. Chem. Phys.*, 2006, **8**, 1057–1065.

13 All computations were performed with the ORCA program package, Version 4.2.1: (*a*) F. Neese, *Wiley Interdiscip. Rev.: Comput. Mol. Sci.*, 2012, **2**, 73–78; (*b*) F. Neese, *Wiley Interdiscip. Rev.: Comput. Mol. Sci.*, 2018, **8**, e1327.

14 A. V. Marenich, C. J. Cramer and D. G. Truhlar, *J. Phys. Chem. B*, 2009, **113**, 6378–6396.

15 A. Music, A. N. Baumann, P. Spieß, A. Plantefol, T. C. Jagau and D. Didier, *J. Am. Chem. Soc.*, 2020, **142**, 4341–4348.